PKU–Zurich PhD Summer School

on Machine Learning for Macroeconomics and Finance

Reading List

Beijing, China · July 6–10, 2026

Before the Summer School (Pre-Reading & Pre-Coding)

Everyone should complete these two items before arriving in Beijing.

- **Reading — heterogeneous-agent models.** Dirk Krueger, *An Introduction to Macroeconomics with Household Heterogeneity* (lecture notes), 2025. [link]

  Read **Chapter 6, "The SIM in General Equilibrium,"** carefully; use **Chapter 4, "The SIM in Partial Equilibrium,"** as a background reference.

- **Coding — neural networks for economic models.** Simon Scheidegger, *Deep Equilibrium Nets: Brock–Mirman tutorials* (Jupyter notebooks).

  Read and run both notebooks: Notebook 1 — deterministic Brock–Mirman, and Notebook 2 — stochastic Brock–Mirman.

Day 1 — Deep Learning and Heterogeneous-Agent Basics

Serguei Maliar (Santa Clara University)

- Maliar, Maliar & Winant (2021). *Deep learning for solving dynamic economic models.* Journal of Monetary Economics 122, 76–101. [link]

- Lepetyuk, Maliar & Maliar (2020). *When the U.S. catches a cold, Canada sneezes: A lower-bound tale told by deep learning.* Journal of Economic Dynamics and Control 117, 103926. [link]

- Maliar & Maliar (2021). *Deep learning classification: Modeling discrete labor choice.* Journal of Economic Dynamics and Control 135, 104295. [link]

Recommended background

- Goodfellow, Bengio & Courville (2016). *Deep Learning.* MIT Press (full text free online). [link]

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving.* Quarterly Journal of Economics 109(3), 659–684. [link]

- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy.* Journal of Political Economy 106(5), 867–896. [link]

Day 2 — Deep Equilibrium Nets

Simon Scheidegger (HEC Lausanne) and Felix Kübler (University of Zurich)

- Scheidegger (2026). *Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance.* arXiv:2605.14493 (lecture notes, with TensorFlow / PyTorch notebooks). [link]

- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets.* International Economic Review 63(4), 1471–1525. Code: github.com/sischei/DeepEquilibriumNets. [link]

- Scheidegger & Bilionis (2019). *Machine learning for high-dimensional dynamic stochastic economies.* Journal of Computational Science 33, 68–82. [link]

- Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing.* Journal of Financial Economics 177, 104222. [link]

- Friedl, Kübler, Scheidegger & Usui (2021). *Deep Uncertainty Quantification: With an Application to Integrated Assessment Models.* Working paper, University of Lausanne. [link]

- Rasmussen & Williams (2006). *Gaussian Processes for Machine Learning.* MIT Press (full text free online). [link]

Day 3 — DeepHAM and Structural Reinforcement Learning

Yucheng Yang (University of Zurich), Benjamin Moll (LSE)

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks.* Quantitative Economics 17(2), 297–341. Code: github.com/frankhan91/DeepHAM. [link]

- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics.* arXiv:2512.18892 (code coming soon). [link]

- Wibault, Forkel, Towers, Wibault, Duque, Whittle, Schaab, Yang, Wang, Osborne, Moll & Foerster (2026). *Recurrent Structural Policy Gradient for Partially Observable Mean Field Games.* International Conference on Machine Learning (ICML), 2026. Code: github.com/CWibault/mfax. [link]

Recommended background

- Moll (2025). *The Trouble with Rational Expectations in Heterogeneous Agent Models: A Challenge for Macroeconomics.* The Economic Journal (forthcoming); Economic Journal Lecture, RES 2024. [link]

Day 4 — Deep Learning and Continuous-Time Macro-Finance

Goutham Gopalakrishna (University of Toronto, Rotman)

- Gopalakrishna & Wu (2021). *ALIENs for Continuous Time Economies.* Swiss Finance Institute Research Paper No. 21-34. [link]

- Raissi, Perdikaris & Karniadakis (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations.* Journal of Computational Physics 378, 686–707. [link]

Recommended background

- Brunnermeier & Sannikov (2014). *A Macroeconomic Model with a Financial Sector.* American Economic Review 104(2), 379–421. [link]

- Achdou, Han, Lasry, Lions & Moll (2022). *Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*. Review of Economic Studies 89(1), 45–86. [link]

Day 5 — Deep Learning for Macro-Finance and Beyond

Jonathan Payne (Princeton University)

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726 (forthcoming, JPE: Macroeconomics). [link]

- Payne, Rebei & Yang (2025). *Deep Learning for Search and Matching Models*. Swiss Finance Institute Research Paper No. 25-05. [link]

- Gopalakrishna, Gu & Payne (2026). *Institutional Asset Pricing with Segmentation and Household Heterogeneity*. SSRN Working Paper. [link]